



HiLCoE

School of Computer Science and Technology

FLEET MANAGEMENT SYSTEM

SOFTWARE REQUIREMENTS SPECIFICATION (SRS) DOCUMENT

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Chapter 3: Software Requirements Specification (SRS)

3.1 Introduction

3.1.1 Purpose

The purpose of this document is to define, completely and unambiguously, what the Fleet Management System (FMS) must do and how well it must do it so that the development team, the project advisor, and anyone evaluating the finished product are working from the same understanding of the system's capabilities and limits.

This document defines, completely and unambiguously, what the Fleet Management System (FMS) must do and the level of performance required so the development team, project advisor, and evaluators share a single, testable understanding of the system's scope and limits. Ethiopian fleet operators today rely on paper records, scattered documents, spreadsheets, and limited GPS add-ons, producing recurring operational gaps: maintenance histories are hard to retrieve, fuel use cannot be reliably audited, dispatching is based on memory rather than live availability, and owners lack a consolidated view of fleet efficiency.

The FMS closes that gap with a single, web-based platform that unifies driver and vehicle records, trip and route management, fuel tracking, maintenance, alerting, and fleet analytics, including a live Fleet Command Center and AI-driven predictive maintenance. It will be delivered as a fully working, production-capable application every capability described, from real-time GPS and live alerting to analytics and predictive maintenance, is intended to operate end-to-end on real data. This SRS is the authoritative reference for building, reviewing, and judging the final system.

This SRS is the reference point the team will build against, the project advisor will review design decisions against, and the evaluation panel will judge whether the final, delivered system does what it was scoped to do.

3.2 Current System (As-Is)

3.2.1 Organizations Currently Managing Fleets in Ethiopia

Sheger Mass Transport Service Enterprise (SMTSE) is one of two public bus transport providers in Addis Ababa. Around 2020–2021, a maintenance technician at SMTSE began digitizing the enterprise's paper- and Word-based maintenance records into a searchable format; this grew, over time and with IT support, into a web-based centralized system covering Maintenance, Fuel, Inventory (spare parts), Tracking, Vehicles, and Administration (HR/depot information). The system reuses the Mobile Digital Video Recording (MDVR) units already installed in SMTSE's buses to add GPS tracking rather than purchasing new hardware, and runs on modest, largely self-provisioned infrastructure. A companion public trip-planning app, requested by the Addis Ababa Transport Bureau, exposes route, fare, and real-time bus location data drawn from the same system.

Anbessa City Bus Service Enterprise (ACBSE), the other public operator, is a much larger and older institution state-owned, founded in 1945, operating over 1,000 buses on roughly 190+ routes and carrying over a million passengers a day. Public disclosures about ACBSE's internal systems are limited, and it has been described as having close to 40% of its fleet out of service at any given time due to technical issues consistent with a fleet still largely dependent on reactive maintenance rather than a centralized, preventive system. SMTSE and ACBSE are in the process of merging into a single "Addis Ababa City Bus Service Enterprise," at which point ACBSE's vehicle and route data is expected to be folded into SMTSE's existing digital system.

Beyond these two public operators, most private bus, logistics, and trucking companies in Ethiopia do not build custom software at all. Where any digitization exists, it typically comes from local GPS-tracking and telematics vendors (e.g., EthioTrack, Kayu Tech, MKTY), who install a tracking device and provide a web/mobile portal for location, speed, and basic fuel-sensor monitoring but not an integrated fleet-management platform covering maintenance, driver performance, or predictive analytics.

3.2.2 How Fleet Management Is Currently Done

Three patterns describe the current state across these organizations:

1. **Paper/manual (most common, especially smaller private operators):** vehicle, driver, fuel, and maintenance records are kept on paper or in Word/Excel files, reconciled and reported by hand, with no linkage between records that describe the same vehicle.
2. **Vendor GPS add-on (some private/commercial fleets):** a third-party device and portal report location and possibly speed/fuel-sensor data, layered on top of otherwise-manual recordkeeping for maintenance and driver management the tracking system and the paper-based maintenance/fuel system never talk to each other.
3. **In-house digitized (SMTSE, the exception):** a custom-built, modular web system was grown organically from a single technician's script into a multi-department platform, but it remains a self-hosted, internally maintained system rather than a commercially available product other operators can adopt.

None of the three patterns currently combines predictive maintenance, fuel-anomaly detection, driver scoring, and a live command-center view in one accessible, adoptable product, the gap this project addresses.

3.2.3 Stakeholders and Their Roles and Interests

Fleet / Operations Manager

- *Role today:* Oversees vehicle assignment, monitors overall fleet performance, and is the point of escalation when something goes wrong (a vehicle breaks down, a route falls behind schedule). In smaller organizations this role is combined with dispatching; in larger ones (SMTSE, ACBSE) it is a distinct management layer above day-to-day dispatch.
- *Current pain points:* Has no consolidated, real-time view of the fleet and must ask multiple people (dispatchers, technicians, finance) to compile even a basic weekly report on utilization or cost. Decisions about which vehicles to retire, repair, or invest in are made from incomplete, delayed information.
- *Interest in a new system:* Wants a single dashboard that answers "how is the fleet doing right now, and where is money being lost?" without needing to ask staff to compile it manually.

Dispatcher

- *Role today:* Assigns drivers to vehicles and routes daily, and is the first point of contact when a driver calls in with a delay, breakdown, or other issue.
- *Current pain points:* Assignment decisions rely entirely on memory or a whiteboard; there's no systematic check of driver license validity, recent duty hours, or vehicle maintenance status at the moment of assignment, so conflicts (assigning a vehicle that's actually in the workshop) do happen.
- *Interest in a new system:* Wants to see, at a glance, which drivers and vehicles are genuinely available and roadworthy before making an assignment, and wants live visibility into a trip once it's underway rather than waiting for a phone call.

Driver

- *Role today:* Operates the assigned vehicle, is expected to report faults verbally, and where a paper trip sheet or logbook exists records basic trip details after the fact.
- *Current pain points:* Has no easy way to report a fault, log fuel, or confirm today's assignment other than a phone call or a face-to-face conversation at the depot; performance/attendance records that affect them are kept by supervisors and not visible to the driver themselves.
- *Interest in a new system:* Wants a simple, fast way to confirm assignments, log fuel purchases, and submit inspection/incident reports without paperwork or waiting to speak to a supervisor.

Maintenance Technician / Mechanic

- *Role today:* Diagnoses and repairs vehicles brought into the workshop, and is generally the person originally responsible (as at SMTSE) for whatever informal recordkeeping system exists.
- *Current pain points:* Cannot quickly retrieve a vehicle's full service history records may be on paper, in a Word file, or simply not written down anywhere retrievable which slows diagnosis and can lead to repeating work or missing a pattern of recurring failure.
- *Interest in a new system:* Wants to pull up a vehicle's complete history by plate number in seconds, and wants advance warning (a predictive alert) before a vehicle fails, rather than only finding out when it breaks down.

Depot Administrator (HR & Records)

- *Role today:* Manages driver personnel files, attendance records, and depot-level administrative tasks (staffing, scheduling of technicians).
- *Current pain points:* Personnel and attendance records are paper-based and not linked to performance or scheduling data, making it hard to answer even simple questions like "how many days has this driver been absent this month?" without manually reviewing files.
- *Interest in a new system:* Wants attendance, licensing, and personnel data centralized and searchable, with automatic reminders for license/document renewals rather than relying on memory.

Fuel / Finance Clerk

- *Role today:* Collects fuel receipts, manually enters them into a ledger or spreadsheet, and periodically (usually monthly) reconciles fuel expenditure against vehicle mileage.
- *Current pain points:* Reconciliation is manual, periodic, and error-prone; discrepancies caused by theft or leakage are typically discovered weeks after the fact, if at all, because there is no continuous, automated comparison of fuel consumed against distance traveled.
- *Interest in a new system:* Wants fuel entries automatically tied to mileage and flagged immediately if a vehicle's consumption looks abnormal, instead of discovering a problem at month-end.

Fleet Owner / General Manager / Executive Management

- *Role today:* Sets budgets, approves major expenditures (vehicle purchases, large repairs), and is ultimately accountable for the organization's operating costs and service quality.
- *Current pain points:* Must rely on ad hoc, manually compiled reports to understand cost trends, utilization, and downtime; there is no standing dashboard, so strategic decisions (fleet expansion, retiring old vehicles) are made on incomplete or stale data.
- *Interest in a new system:* Wants a trustworthy, always-current view of fleet cost, utilization, and health to support investment and budgeting decisions.

IT / Technical Support Staff

- *Role today:* Where any digital system exists at all (as at SMTSE), a small technical staff, sometimes a single person, as was the case originally builds and maintains it alongside other duties, without dedicated tooling or budget.
- *Current pain points:* Building and maintaining even a basic system is done without institutional support, budget, or dedicated staff time, which limits how far any in-house digitization effort can realistically go.
- *Interest in a new system:* Wants a system that does not require ongoing custom development to maintain configuration and support, not software engineering, should be the ongoing task.

Regulator (e.g., Addis Ababa Transport Bureau, INSA)

- *Role today:* Oversees public transport service levels and, where public-facing data is involved (such as SMTSE's trip-planning app), must approve its release.
- *Current pain points:* Has limited standardized data from operators to support route planning or service-level oversight, since most operators do not maintain digital, shareable records.
- *Interest in a new system:* Wants access to reliable route/service data for planning purposes, within an appropriate approval and data-governance process.

Passengers / Public (indirect)

- *Role today:* Riders of public transport services, generally without access to real-time information about bus location or expected arrival.
- *Current pain points:* Cannot plan around real-time vehicle location or delays; must estimate arrival based on a fixed schedule that may not reflect actual conditions.
- *Interest in a new system:* Indirectly benefits from more reliable service resulting from better fleet management, and potentially from public-facing trip information if an organization chooses to expose it (as SMTSE has done).

3.2.4 Workflow Description

(a) Vehicle Maintenance Workflow

1. A driver notices unusual vehicle behavior (noise, warning light, reduced performance) during a trip or a pre-trip check.
2. The driver reports the issue verbally or by phone to the depot supervisor/dispatcher there is no formal digital fault-reporting channel.
3. The dispatcher or supervisor judges whether the vehicle can complete its current trip or must return to the depot immediately.
4. If the vehicle returns, it is logged into the workshop queue, typically by writing the plate number and reported issue on a whiteboard or in a logbook.
5. A technician is assigned to inspect the vehicle, generally based on availability rather than the technician's specific familiarity with that vehicle's history.
6. The technician attempts to locate the vehicle's prior maintenance records, which may be spread across paper folders, a Word document, or may not exist in any organized form at all.
7. The technician performs a diagnostic inspection relying on physical inspection and experience, since no onboard diagnostic-code history is captured or stored anywhere.
8. Once the fault is identified, the technician checks whether the required spare part is in stock by physically looking in the parts store or asking the storekeeper if there is no digital inventory system tied to maintenance needs.
9. If the part is unavailable, a purchase request is raised manually (a paper requisition or a verbal request), and the vehicle remains out of service until the part arrives, with no formal tracking or escalation of the delay.
10. Once the part is available, the technician performs the repair.
11. The technician records the work performed date, issue description, part used in a logbook or Word document, without any automatic link to the vehicle's mileage, cost history, or a broader health assessment.
12. The vehicle is returned to service, and the dispatcher is informed verbally that it is available again.
13. There is no automatic recalculation of the next service due date/mileage; this depends on a staff member remembering to check or estimate it from the last recorded service.

(b) Fuel Purchase and Reconciliation Workflow

1. A driver identifies the need to refuel, typically from a visual fuel-gauge check or a fixed refueling schedule rather than a calculated remaining-range estimate.

2. The driver stops at an approved fuel station (or a depot's own fuel pump) and refuels the vehicle.
3. The driver obtains a paper receipt, or, at a depot pump, the amount is recorded by the pump attendant in a logbook.
4. The driver retains the receipt until returning to the depot, a window in which receipts can be lost, delayed, or altered.
5. On return, the driver hands the receipt to the depot administrator or finance clerk.
6. The finance clerk manually transcribes the receipt details (date, amount, liters, vehicle plate) into a ledger or spreadsheet.
7. At the end of a reporting period (commonly monthly), the finance clerk totals fuel expenditure per vehicle from these manual entries.
8. Reconciliation against the vehicle's actual mileage for the period, where it happens at all, is done by manually pulling odometer readings from paper records and calculating consumption by hand.
9. Because both fuel and mileage records are manually entered and only reconciled periodically, discrepancies caused by theft, leakage, or misreporting are typically detected weeks after the fact, if at all.
10. There is no automated alert when a vehicle's fuel consumption deviates from its historical norm; deviations are noticed only if a staff member happens to compare records and spot the anomaly.

(c) Trip Dispatch Workflow

1. A transport requirement arises (a fixed public bus route or a scheduled transport order).
2. The dispatcher reviews, from memory or a whiteboard/logbook, which drivers are on duty and which vehicles are not flagged as under repair.
3. The dispatcher manually assigns a driver and vehicle, without a systematic check of license validity, recent performance history, or duty-hour limits.
4. The assignment is communicated verbally or by phone/radio to the driver.
5. During the trip, the driver's location is known only through radio/phone check-ins, or, where a vendor GPS tracker exists, through a separate portal not connected to the trip record itself.
6. If a delay or deviation occurs, the dispatcher becomes aware only if the driver calls in, or if someone happens to check the separate tracking portal.
7. On completion, the driver reports back verbally or via a paper trip sheet, which is later, if ever, reconciled against fuel or maintenance data.

(d) Driver Onboarding and Attendance Workflow

1. A new driver is hired; identification and license documents are collected in physical, paper form.
2. Driver details are recorded in a personnel file or HR ledger, generally without a digital, searchable record.
3. Daily attendance is recorded via a physical sign-in sheet at the depot or a supervisor's manual roll call.
4. Performance issues (lateness, complaints, accidents) are recorded, if at all, as handwritten notes in the driver's personnel file, disconnected from any performance-scoring mechanism.
5. License-renewal dates are tracked only insofar as a supervisor or the driver personally remembers to check them there is no automatic expiry alert.

(e) Vehicle Registration Workflow

1. A new vehicle is purchased or leased.
2. Registration, insurance, and licensing documents are filed physically.
3. The vehicle is assigned a plate number and added to a paper or spreadsheet asset register.
4. Specification details (make, model, capacity, fuel type) are recorded manually, without systematic linkage to the maintenance or fuel systems above.

(f) Incident/Accident Reporting Workflow

1. An accident or safety incident occurs during a trip.
2. The driver reports the incident verbally, by phone, to the dispatcher or supervisor as soon as possible.
3. A written incident report is completed after the fact, typically once the driver returns to the depot a delay between the event and its documented record.
4. The report is filed in the driver's personnel folder or a separate incident log, generally without cross-referencing the vehicle's own maintenance/damage history.
5. Follow-up (insurance claims, disciplinary review, safety-pattern analysis) is handled case by case, since there is no aggregated view of incidents across drivers or vehicles.

3.2.5 Business Rules Currently Operating

Vehicle Assignment / Dispatch

- A vehicle undergoing repair should not be assigned to a trip, but this is enforced only by a supervisor's memory or a whiteboard note; there is no system-enforced restriction, so assignment errors do occur.
- Drivers are typically assigned to a fixed vehicle or route on a semi-permanent basis (especially in public transit), rather than dynamically reassigned daily, since dynamic reassignment would require real-time availability information the organization does not have digitally.

Maintenance

- Maintenance is triggered by either a fixed interval (kilometers or months) or a reported fault, whichever comes first but because mileage tracking is manual, the interval-based trigger is frequently missed or applied late.
- A vehicle is not returned to service until a technician verbally confirms the repair is complete; there is no formal sign-off record beyond a logbook entry.
- Spare-parts procurement generally requires supervisor or purchasing-officer approval above an informally understood (not systematically enforced) cost threshold.

Fuel

- Only fuel purchases with a retained receipt are considered valid for reimbursement/accounting but retention itself is not verifiable digitally, so lost receipts are handled case-by-case, often as an unreconciled write-off.
- Fuel reconciliation, when performed, happens on a fixed periodic cycle (commonly monthly) rather than continuously, meaning anomalies are structurally detected late by design, not merely by oversight.

Attendance and Driver Performance

- A driver's absence must be reported to a supervisor in advance where possible; same-day absence is handled ad hoc.
- Performance-related decisions (route reassignment, disciplinary action) are made by supervisor judgment, since no standardized, data-based performance score exists in the current system.

Documentation and Compliance

- Vehicles must carry valid registration and insurance documents at all times, but responsibility for tracking renewal dates rests entirely on individual staff or drivers rather than being centrally monitored.
- Public release of any tracking or route information, where a digital tracking capability exists at all (as at SMTSE), requires internal approval and, in that specific case, clearance from a national regulatory body (INSA) before public deployment.

Data Access

- Access to maintenance and fuel records is governed only by physical or file access — anyone with access to the depot office's paper files or shared Word/Excel documents can view or, in principle, alter them, since there is no access-control mechanism in place.

3.2.6 Technologies Currently Used, by Task

Task	Current Technology	Notes/Limitations
Vehicle/driver records	Paper files, Word documents, Excel spreadsheets	No cross-linking between record types; searching is manual
Location tracking	None (most operators); third-party GPS devices and vendor portals (EthioTrack, Kayu Tech, MKTY)	Reports location only; not connected to predictive maintenance, fuel, or dispatch data

Task	Current Technology	Notes/Limitations
Maintenance history	Paper/Word records; SMTSE's custom Maintenance module (the only known integrated example)	Outside SMTSE, retrieval is slow and history is often incomplete
Fuel monitoring	Paper receipts and manual ledger/spreadsheet entry; some vendors offer fuel-level sensor add-ons	Reconciliation is periodic and manual; anomalies detected late
Mapping	Where used at all, low-cost/open options are preferred over paid APIs (SMTSE uses OpenStreetMap rather than Google Maps)	Cost-driven choice reflecting constrained IT budgets
Hosting/infrastructure	Small-scale, self-hosted, on-premises servers where any system exists (SMTSE started on a basic desktop-class machine; depots initially lacked internet connectivity)	No cloud-hosted, multi-tenant option identified in the local market for integrated fleet management
Attendance/HR	Physical sign-in sheets, personnel folders	No link to scheduling or performance systems
Predictive analytics	Not present in any reviewed current system	Entirely reactive maintenance model

3.3 Proposed System Overview

3.3.1 System Context

The FMS sits between its users and a set of surrounding systems/data sources:

- **Users** (via web browser or mobile app): Administrator, Fleet Manager, Dispatcher, Driver, Maintenance Technician, Depot Administrator, Fuel/Finance Clerk, Compliance/Safety Officer, Fleet Owner/Executive.
- **GPS/Telematics Data Source**: a connected GPS/telematics hardware device or a driver-side mobile application, providing real-time location, speed, and vehicle-diagnostic data.
- **Notification Gateway**: real email, SMS, and push-notification providers.
- **Database**: persistent store for all fleet, trip, fuel, maintenance, and user data.
- **Mapping/Traffic/Weather Services**: render vehicle positions and routes and provide live traffic/weather overlays; an open-source or cost-efficient map layer is preferred, consistent with practice observed at SMTSE.

3.3.2 Product **Features** (High-Level)

1. Driver & Vehicle Management
2. Trip & Route Management
3. Fuel Tracking
4. Alerts & Notifications
5. Fleet Analytics Dashboard
6. Fleet Command Center
7. AI-Based Predictive Maintenance
8. EV Fleet Management (for organizations operating electric vehicles) — all governed by a cross-cutting Role-Based Access Control (RBAC) layer.

3.3.3 User Classes and Characteristics

User Class	Technical Proficiency	Frequency of Use	Primary Access
System Administrator	High	Occasional (setup/maintenance of accounts)	Full system configuration
Fleet Manager	Moderate	Daily/weekly	Analytics, command center, approvals
Dispatcher	Moderate	Daily, continuous during shifts	Trips, command center, alerts
Driver	Low–Moderate (mobile-first)	Daily, brief sessions	Own trips, fuel logging, own alerts
Maintenance Technician	Moderate	Daily/as-needed	Maintenance records, predictive alerts
Depot Administrator	Moderate	Daily	Driver/vehicle records, attendance
Fuel/Finance Clerk	Moderate	Daily/weekly	Fuel records, reconciliation reports
Compliance/Safety Officer	Moderate	Weekly/as-needed	DVIR, incident reports, license compliance
Fleet Owner/Executive	Low–Moderate	Weekly/monthly	Read-mostly dashboards and reports

3.3.4 Operating Environment

Web-based, browser-independent client (desktop and mobile browsers), with a companion mobile-friendly interface for drivers; backend and database platform to be finalized in the System Design

Document (Chapter 4); deployable to a cloud host or an on-premises server, reflecting that many target organizations as seen with SMTSE begin with modest, low-cost infrastructure.

3.3.5 Design and Implementation Constraints

- Must integrate with a real source of GPS/vehicle telemetry either a connected telematics/OBD device or a driver-side mobile application's GPS sensor.
- Must operate on a limited-budget hosting setup; open-source mapping and libraries are preferred over paid APIs where feasible.
- Must be delivered within the senior-project academic timeline by a four-person team.
- RBAC is a mandatory constraint on every module no feature may bypass role checks.

3.3.7 Assumptions and Dependencies

- Assumes reliable internet connectivity at the point of deployment (a real constraint observed at SMTSE's depots, which initially had none).
- Assumes the deploying organization can provide, or the team can supply, a GPS-capable device or use drivers' mobile phones as the location source.
- Depends on open-source mapping and charting libraries, and on a real third-party email/SMS/push notification provider.
- Depends on the availability of historical or synthetic training data to validate the predictive-maintenance model's accuracy before go-live.

3.3.8 Risks and External Factors

- **Data risk:** limited real historical fleet data exists to validate predictive-maintenance accuracy at launch.
- **Adoption risk:** as seen at SMTSE, staff can be reluctant to adopt a new system; training and change management, while outside this SRS, affect real-world success.
- **Regulatory risk:** if the system is deployed to handle real public-transport data, approval from bodies such as Ethiopia's Information Network Security Administration (INSA) may be required before public data release, as it was for SMTSE's trip-planning app.
- **Infrastructure risk:** target organizations may lack reliable internet/power at all depots, which the system's error-handling behavior (§3.12) must account for.

3.4 Actors and Business Rules

3.4.1 Actors

Actor	Type	Description
System Administrator	Human	Configures the system, manages accounts/roles, maintains master data
Fleet Manager	Human	Oversees fleet-wide performance, budgets, and approves major exceptions
Dispatcher	Human	Assigns drivers/vehicles, schedules trips, monitors the command center
Driver	Human	Views trips, submits DVIR/incident reports, logs fuel purchases, receives alerts
Maintenance Technician	Human	Actions maintenance alerts, logs repairs, updates vehicle health
Depot Administrator	Human	Manages driver personnel records, attendance, and depot-level HR data
Fuel/Finance Clerk	Human	Reviews and reconciles fuel records, manages cost reporting
Compliance/Safety Officer	Human	Reviews DVIR and incident reports, monitors license/document compliance
Fleet Owner/Executive	Human	Views dashboards/KPI reports (read-mostly)
Technical/IT Administrator	Human	Monitors system health, backups, and integration configuration

Actor	Type	Description
GPS/Telematics Device or Mobile App	Hardware/Software	Supplies real-time location, speed, and diagnostic data
Notification Gateway	Software	Real email/SMS/push delivery provider
Mapping/Traffic Service	Software	Supplies map rendering and live traffic data
Weather Service	Software	Supplies live weather data for command-center overlays
EV Charging Station System	Hardware/Software	Reports charging status and energy data for EV-equipped vehicles

3.4.2 Business Rules (Proposed System)

- BR-1: A driver cannot be assigned to two overlapping trips.
- BR-2: A vehicle flagged for active maintenance cannot be dispatched on a trip.
- BR-3: Only a System Administrator may create an account with a role above Driver.
- BR-4: A fuel entry cannot be edited more than 48 hours after submission (to preserve reconciliation integrity).
- BR-5: A maintenance-due alert must be generated automatically when a vehicle reaches its configured mileage/engine-hour/time threshold it must not depend on manual review.
- BR-6: A predictive-maintenance alert above a configurable risk threshold must be routed to a Maintenance Technician, not only to the Fleet Manager.
- BR-7: Any change to a vehicle's maintenance or fuel history must be attributed to the user who made it (see §3.13, Logging).
- BR-8: A driver whose license or required document has expired must be flagged as ineligible for new trip assignment until the document is updated.
- BR-9: An incident/accident report must be linked to both the driver and the vehicle involved, so it appears in both records.

3.5 Functional Requirements

3.5.1 Requirement Identification and Priority Scheme

Each functional requirement is identified as **FR-#**, grouped by module, and assigned a MoSCoW priority (Must/Should/Could). Requirements trace back to the nine specific objectives defined in the Proposal (1.3.2). Non-functional requirements follow the same ID scheme (**NFR-#**) in 3.8.

3.5.2 Functional Requirements by Module

Driver & Vehicle Management

ID	Requirement	Actor(s)	Priority
FR-1	The system shall maintain a profile record for each vehicle (specifications, licensing, identification).	Admin	Must
FR-2	The system shall maintain a profile record for each driver, including license information and qualifications.	Admin	Must
FR-3	The system shall allow a Fleet Manager or Dispatcher to assign a driver to a vehicle based on availability and qualification.	Dispatcher	Must
FR-4	The system shall record driver attendance and compute a performance score.	Depot Admin	Must
FR-5	The system shall capture DVIR and accident report data submitted by a Driver.	Driver	Must

Trip & Route Management

ID	Requirement	Actor(s)	Priority
FR-6	The system shall allow a Dispatcher to schedule and assign a trip to a driver and vehicle.	Dispatcher	Must

FR-7	The system shall optimize a route based on distance and configurable parameters.	Dispatcher	Should
FR-8	The system shall track and reconcile trip mileage against vehicle odometer records.	System	Must
FR-9	The system shall monitor ETA and flag delays.	System	Should
FR-10	The system shall provide real-time GPS trip visualization on a map, using live location data from a connected device or driver mobile app.	Dispatcher, Driver	Must

Fuel Tracking

ID	Requirement	Actor(s)	Priority
FR-11	The system shall allow a Driver to log a fuel purchase against a vehicle.	Driver	Must
FR-12	The system shall reconcile recorded fuel consumption against mileage to compute cost-per-kilometer.	System	Must
FR-13	The system shall detect and flag abnormal fuel-usage patterns indicative of theft or leakage.	System	Must

Alerts & Notifications

ID	Requirement	Actor(s)	Priority
FR-14	The system shall generate a maintenance-due alert based on mileage, engine hours, or elapsed time.	System	Must

FR-15	The system shall generate an alert when a driver's license or vehicle document is approaching expiry.	System	Must
FR-16	The system shall generate real-time alerts for speeding, idling, and unauthorized usage based on live vehicle telemetry.	System	Must
FR-17	The system shall deliver alerts through real email, SMS, and push-notification channels.	System	Must

Analytics Dashboard

ID	Requirement	Actor(s)	Priority
FR-18	The system shall calculate and display fleet utilization rate, distance traveled, and downtime.	Fleet Manager, Owner	Must
FR-19	The system shall calculate and display fuel and maintenance cost trends over a configurable period.	Fleet Manager, Owner	Must
FR-20	The system shall generate a chart-based driver ranking report.	Fleet Manager	Should
FR-21	The system shall generate a vehicle health score report.	Fleet Manager, Technician	Must

Fleet Command Center

ID	Requirement	Actor(s)	Priority
FR-22	The system shall display a live map of active vehicle positions using real GPS/telematics data.	Dispatcher	Must

FR-23	The system shall display real-time trip and driver status alongside the map.	Dispatcher	Must
FR-24	The system shall display live traffic and weather overlays on the map.	Dispatcher	Should

Predictive Maintenance

ID	Requirement	Actor(s)	Priority
FR-25	The system shall forecast upcoming maintenance needs from mileage and engine-hour data.	System	Must
FR-26	The system shall compute a standardized vehicle health score incorporating repair history and efficiency trends.	System	Must
FR-27	The system shall notify a Maintenance Technician when a vehicle's predicted failure risk exceeds a configurable threshold.	System, Technician	Must

EV Fleet Management (for organizations operating electric vehicles)

ID	Requirement	Actor(s)	Priority
FR-28	The system shall monitor battery state-of-charge and estimated range for EV-equipped vehicles.	System	Should
FR-29	The system shall manage and display charging schedules for EV-equipped vehicles.	Fleet Manager	Should
FR-30	The system shall track energy consumption, cost-per-kWh, and battery health for EV-equipped vehicles.	System	Should

Access Control (cross-cutting)

ID	Requirement	Actor(s)	Priority
FR-31	The system shall restrict each user's visible data and available actions according to their assigned role.	System	Must
FR-32	The system shall require Administrator approval before granting any role above Driver.	Admin	Must

3.5.3 User Stories

System Administrator

- As a System Administrator, I want to create role-based accounts, so that users only see data relevant to their role.
- As a System Administrator, I want to register new vehicles and depots, so that the fleet database stays current as the organization grows.
- As a System Administrator, I want to deactivate a departed employee's account immediately, so that access is revoked without delay.
- As a System Administrator, I want to configure alert thresholds (e.g., maintenance mileage intervals), so that the system reflects our organization's own maintenance policy.

Fleet Manager

- As a Fleet Manager, I want to see fuel and maintenance cost trends on a dashboard, so that I can identify vehicles or routes driving up costs.
- As a Fleet Manager, I want to see a driver ranking report, so that I can recognize strong performers and identify who needs coaching.
- As a Fleet Manager, I want to approve exceptions to standard vehicle assignment rules, so that operational judgment can override the default logic when needed.
- As a Fleet Manager, I want to view utilization and downtime by vehicle, so that I can decide which vehicles to retire or replace.

Dispatcher

- As a Dispatcher, I want to see a live map of active vehicles and their status, so that I can respond quickly to delays or breakdowns.
- As a Dispatcher, I want the system to prevent me from assigning a vehicle flagged for maintenance, so that I don't accidentally dispatch an unsafe vehicle.
- As a Dispatcher, I want to see each driver's current duty status before assigning a trip, so that I avoid overlapping assignments.
- As a Dispatcher, I want to be notified immediately if a trip falls significantly behind its ETA, so that I can respond before a passenger or customer complains.

Driver

- As a Driver, I want to see my assigned trip and route details on login, so that I know where and when I need to be.
- As a Driver, I want to log a fuel purchase quickly (just plate number and amount), so that I don't have to re-enter information the system already has.
- As a Driver, I want to receive an alert when my vehicle is due for maintenance or my license is close to expiry, so that I can act before it becomes a compliance issue.
- As a Driver, I want to submit a DVIR or incident report from my phone, so that I don't have to wait until I'm back at the depot to file paperwork.
- As a Driver, I want to see my own attendance and performance record, so that I understand how I'm being evaluated.

Maintenance Technician

- As a Maintenance Technician, I want to pull up a vehicle's full service history by plate number, so that I can diagnose issues faster.
- As a Maintenance Technician, I want to receive a predictive-maintenance alert with an estimated risk level, so that I can plan service before a breakdown occurs.
- As a Maintenance Technician, I want to log completed repairs and parts consumed, so that the vehicle's health score and inventory update automatically.

- As a Maintenance Technician, I want to see which vehicles are currently flagged and awaiting service, so that I can prioritize my workload.

Depot Administrator

- As a Depot Administrator, I want to record daily driver attendance digitally, so that I don't need a paper sign-in sheet.
- As a Depot Administrator, I want automatic reminders before a driver's license expires, so that I don't have to track renewal dates manually.
- As a Depot Administrator, I want a searchable personnel record for each driver, so that I can answer HR questions without digging through paper files.

Fuel/Finance Clerk

- As a Fuel/Finance Clerk, I want fuel entries automatically reconciled against mileage, so that I don't have to calculate consumption by hand.
- As a Fuel/Finance Clerk, I want to be alerted immediately when a vehicle's fuel usage looks abnormal, so that I can investigate before month-end.
- As a Fuel/Finance Clerk, I want a monthly cost report per vehicle, so that I can support budgeting without manually compiling receipts.

Compliance/Safety Officer

- As a Compliance/Safety Officer, I want to review all submitted DVIR and incident reports in one place, so that I can spot safety patterns across drivers or vehicles.
- As a Compliance/Safety Officer, I want a list of drivers or vehicles with expiring documents, so that I can follow up before they become non-compliant.

Fleet Owner/Executive

- As a Fleet Owner, I want a single dashboard showing utilization, downtime, and cost trends across the whole fleet, so that I can make budget and investment decisions without asking staff to compile reports manually.

- As a Fleet Owner, I want to see a fleet-wide health score trend over time, so that I can judge whether our maintenance approach is improving or declining.

3.6 Proposed System Models (Use Cases, Object Model, Dynamic Models)

3.6.1 Use Case Diagram (textual)- to be done (Place holder)

- ~~Admin — manages accounts, manages master data~~
- ~~Dispatcher — assigns trip, monitors command center, views analytics~~
- ~~Driver — views trip, logs fuel, submits DVIR~~
- ~~Technician — views maintenance alert, logs repair~~
- ~~Depot Admin — manages attendance, personnel records~~
- ~~Fuel/Finance Clerk — reviews and reconciles fuel records~~
- ~~Compliance Officer — reviews DVIR/incident reports, monitors compliance~~
- ~~Owner — views dashboard~~
- ~~(includes): "Assign Trip" includes "Check Driver Availability"; "Log Fuel Purchase" includes "Reconcile Mileage"~~
- ~~(extends): "Monitor Command Center" extends to "View Traffic/Weather Overlay"~~

3.6.2 Use Case Descriptions (Sample)

UC-1 — Schedule and Assign a Trip

- *Primary Actor:* Dispatcher
- *Preconditions:* At least one available, qualified driver and one roadworthy vehicle exist.
- *Main Flow:* Dispatcher selects a route → system suggests available drivers/vehicles (excluding those under BR-1/BR-2 conflicts) → Dispatcher confirms assignment → system creates the trip and notifies the Driver.
- *Alternate Flow:* If no eligible driver/vehicle exists, system displays a conflict message and suggests the next available option.

- *Postconditions:* Trip is created and visible on the Driver's and Dispatcher's views.

UC-2 — Log a Fuel Purchase

- *Primary Actor:* Driver
- *Preconditions:* Driver is authenticated and assigned to an active vehicle.
- *Main Flow:* Driver enters plate number and fuel amount → system auto-fills vehicle/trip context → system stores the entry and updates cost-per-kilometer (FR-12).
- *Alternate Flow:* If the entered amount deviates sharply from the vehicle's historical pattern, the system flags it for review (FR-13).
- *Postconditions:* Fuel log stored; analytics updated.

UC-3 — Receive and Act on a Predictive Maintenance Alert

- *Primary Actor:* Maintenance Technician
- *Preconditions:* Vehicle has sufficient mileage/engine-hour history for a risk estimate.
- *Main Flow:* System computes risk score → if above threshold (BR-6), alert is routed to Technician → Technician reviews the vehicle's health score and history → Technician schedules/performs service → Technician logs the outcome.
- *Postconditions:* Vehicle health score and maintenance history are updated; alert is marked resolved.

3.6.3 Activity Diagram (textual — Predictive Maintenance Alert Flow)

Vehicle telemetry received → risk score computed → **[decision]** risk $\geq$ threshold? → **No:** continue monitoring → **Yes:** generate alert → route to Technician → Technician reviews history → Technician performs/schedules service → log outcome → update health score → close alert.

3.6.4 Class Diagram (Analysis-Level)

Class	Key Attributes	Key Operations
User	id, name, role, credentials	login(), changeRole()

Class	Key Attributes	Key Operations
Vehicle	plateNumber, type, specs, healthScore	updateHealthScore(), flagForMaintenance()
Driver	licenseNo, qualifications, performanceScore	recordAttendance()
Trip	route, assignedDriver, assignedVehicle, status, ETA	assign(), reconcileMileage()
FuelLog	vehicleId, amount, date, costPerKm	validateAmount(), flagAnomaly()
MaintenanceRecord	vehicleId, date, description, partsUsed	logRepair(), updateInventory()
Alert	type, severity, targetActor, status	route(), resolve()
IncidentReport	driverId, vehicleId, description, date	fileReport(), linkToRecords()
EVBatteryLog (<i>for EV-equipped vehicles</i>)	SoC, range, cyclesCount	estimateRange(), scheduleCharge()

3.6.5 Decision Table (Sample — Predictive Maintenance Risk Routing)

Condition	Rule 1	Rule 2	Rule 3
Risk score	Low	Medium	High
Vehicle currently on active trip?	—	No	Yes
Action	Log only	Schedule routine service	Immediate alert + flag vehicle for maintenance (BR-2)

3.6.6 Sequence Diagram (textual — UC-2, Log a Fuel Purchase) - **place holder**

~~Driver → UI: enter plate number, amount → UI → API: submit(fuelEntry) → API → DB: fetch vehicle history → DB → API: return history → API: compute cost per km, check anomaly (FR-12, FR-13) → API → DB: save entry → API → UI: confirmation → UI → Driver: display success/flag message.~~

3.6.7 Data Flow Diagram — Level 0 (Context, textual) - **place holder**

~~**External entities:** Driver, Dispatcher, Technician, Depot Admin, Fuel Clerk, Compliance Officer, Owner, GPS/Telematics Source, Notification Gateway. **Process 0:** Fleet Management System. **Data stores:** Vehicle/Driver DB, Trip DB, Fuel/Maintenance DB, Alert DB, Incident DB. Flows: Driver → (fuel/DVIR data) → Process 0 → (fuel/maintenance records) → Data stores; GPS/Telematics Source → (location/speed data) → Process 0 → (alerts) → Notification Gateway → (alerts) → Driver/Dispatcher/Technician; Dispatcher → (trip assignment) → Process 0; Owner ← (reports) ← Process 0.~~

3.7 External Interface Requirements

3.7.1 User Interfaces

Responsive, role-based web UI: a Dispatcher/Owner dashboard view, a Driver mobile-first view (trip card, fuel-entry form, DVIR form), a Technician maintenance view (history lookup, repair logging), a Depot Admin view (attendance, personnel), and a Fuel/Finance view (reconciliation, cost reports). Navigation is role-scoped; a user only sees menu items for actions their role permits (per FR-31).

3.7.2 Hardware Interfaces

The system integrates with a GPS-enabled telematics device (e.g., an OBD-II/telematics dongle) where available, or a driver-side mobile application's built-in GPS sensor, to obtain real-time location and vehicle telemetry data. The data-ingestion interface is designed so that different hardware sources can be substituted without changing the underlying data model.

3.7.3 Software Interfaces

- Database interface (CRUD access to Vehicle, Driver, Trip, FuelLog, MaintenanceRecord, Alert, IncidentReport, User entities).
- Mapping library interface (open-source map tiles/rendering, e.g., an OpenStreetMap-based library, to control cost).
- Real notification interface (email/SMS/push provider APIs).
- Authentication/RBAC interface.
- Optional fuel-card/payment-system interface, for organizations that use fuel cards.

3.7.4 Communication Interfaces

HTTPS/REST API for all client-server communication; a WebSocket (or efficient polling) channel for near-real-time command-center updates (live vehicle positions/status).

3.8 Non-Functional Requirements

3.8.1 Performance

- NFR-1: The dashboard shall load within 3 seconds for 95% of requests under normal load.
- NFR-2: API responses for standard queries (e.g., trip list, vehicle profile) shall return within 500 ms under normal load.
- NFR-3: The command center shall support tracking of at least 500 concurrent vehicles without visible lag.

3.8.2 Availability and Reliability

- NFR-4: The system shall target 99% uptime.
- NFR-5: If the GPS/telematics data feed is temporarily unavailable (e.g., due to poor network coverage), the system shall display the vehicle's last known position with a "data stale" indicator rather than failing (see §3.12).

3.8.3 Security and Privacy

- NFR-6: The system shall enforce RBAC on every API endpoint, not only in the UI.
- NFR-7: Passwords shall be stored using a salted hash; no plaintext credentials shall be stored.
- NFR-8: All client-server traffic shall be encrypted via HTTPS.
- NFR-9: Driver behavioral data (location, speed, driving events) shall be accessible only to roles with a legitimate operational need (Dispatcher, Technician, Compliance Officer, Owner).

3.8.4 Usability and Accessibility

- NFR-10: A first-time user in the target role shall be able to complete their primary task (e.g., log a fuel purchase) without external training, using in-app guidance alone.
- NFR-11: The UI shall maintain readable contrast and legible font sizes on both desktop and mobile viewports.

3.8.5 Maintainability / Supportability

- NFR-12: Each of the eight functional modules shall be implemented as a separable component to allow independent modification without full-system redeployment.
- NFR-13: All modules shall include internal code documentation per the conventions defined in Chapter 6.3.

3.8.6 Portability / Compatibility

- NFR-14: The web client shall function correctly on the current versions of Chrome, Firefox, Edge, and Safari.
- NFR-15: The system shall not depend on any operating-system-specific feature on the client side.

3.9 Data Requirements

Key entities: **User** (id, role, credentials), **Vehicle** (plate, specs, health score), **Driver** (license, qualifications, performance score), **Trip** (route, driver, vehicle, status), **FuelLog** (vehicle, amount, date), **MaintenanceRecord** (vehicle, date, description, parts), **Alert** (type, severity, status), **IncidentReport** (driver, vehicle, description, date), and, for EV-equipped fleets, **EVBatteryLog**. Validation rules include:

mileage and fuel amounts must be non-negative; a trip's end time must be after its start time; a maintenance record must reference an existing vehicle. Historical fuel, trip, and maintenance data should be retained for the life of the vehicle record to support analytics and predictive maintenance.

3.10 Regulatory / Standards Compliance

Ethiopia currently has no dedicated GPS-tracking or comprehensive data-protection law, though this is an evolving area; the system should nonetheless follow general good practice (role-restricted access to location/behavioral data, per NFR-9) in anticipation of future regulation. Where real telematics hardware is integrated, the system should align with relevant industry standards referenced in the literature review — SAE J1939 for heavy-duty vehicle diagnostics and ISO/TS 15143-3 for telematics data exchange — to ease integration with real vehicle hardware. Any future public data release (e.g., a passenger-facing app) should anticipate a regulatory approval step, as was required of SMTSE's trip-planning app before public launch.

3.11 Error Handling and Fault Responses

- If the GPS/telematics data feed is temporarily unavailable, the system shall show the last known vehicle position with a "data stale" indicator (NFR-5) rather than a blank map.
- If an alert cannot be delivered through a given channel (e.g., SMS delivery failure), the system shall retry once and then fall back to an in-app notification.
- If a data submission fails validation (e.g., negative mileage), the system shall reject it with a specific, actionable error message rather than a generic failure.
- If two users attempt to edit the same record concurrently, the system shall apply a last-write-wins rule and log the conflict for audit (§3.13).

3.12 Logging, Audit, and Monitoring

The system shall log: all authentication attempts (success/failure), all role/account changes, all edits to fuel and maintenance records (with the editing user's identity, per BR-7), and all system errors. Logs shall be retained for a configurable period and shall be accessible only to the System Administrator role. This log is the primary control against the fuel-theft/record-tampering risk identified in the Problem Description (Chapter 1, §1.2).

3.13 User Interface (Mock-ups, Navigational Flow)

Screen	Primary Actor(s)	Purpose	Navigates To
Login	All	Authenticate and route to role-specific home	Role home screen
Fleet Manager Dashboard	Fleet Manager, Owner	Utilization, cost trends, driver ranking, health scores	Vehicle Profile, Driver Profile
Dispatcher Home	Dispatcher	Trip list, quick-assign, alert summary	Command Center, Trip Detail
Command Center	Dispatcher	Live map, vehicle/driver status	Trip Detail
Driver Home	Driver	Today's trip, quick fuel-log entry, DVIR form	Trip Detail, Fuel Log
Maintenance Console	Technician	Alert queue, vehicle history lookup, repair logging	Vehicle Profile
Depot Admin Console	Depot Administrator	Attendance, personnel records	Driver Profile
Fuel/Finance Console	Fuel/Finance Clerk	Fuel entries, reconciliation, cost reports	Vehicle Profile
Compliance Console	Compliance Officer	DVIR/incident review, document-expiry list	Driver Profile, Vehicle Profile

Screen	Primary Actor(s)	Purpose	Navigates To
Admin Console	Administrator	Account/role management, master data	—

Detailed screen mock-ups and a navigable prototype are planned as part of Chapter 3.7/Chapter 5 deliverables; the table above defines the navigation skeleton they should follow.